

BHUPINDER K

Phone: 469- 305- 1480

Gmail: bk3345439@gmail.com

LinkedIn: <http://linkedin.com/in/bhupinder-k-326a97246>

CERTIFICATIONS

Microsoft Certified: DevOps Engineer Expert:

https://www.credly.com/badges/ffc9b8c1-261d-44a7-bc4b-1d80fd94f9d3/public_url

AZ-400: Designing and Implementing Microsoft DevOps Solutions:

https://www.credly.com/badges/c48c4177-3007-4847-bef0-8e486a5b0bd1/public_url

Microsoft Certified: Azure Administrator Associate:

https://www.credly.com/badges/50b2dab7-c68d-440a-a1d3-3a48c1a27d5d/public_url

PROFESSIONAL SUMMARY

Versatile and certified DevOps expert with 12+ years of experience spanning **Azure**, **AWS**, **GCP** and **Linux** environments, progressing through critical roles such as **Azure DevOps Architect**, **Senior DevOps Engineer**, and **Infrastructure Engineer**. Skilled in architecting and automating cloud infrastructure, implementing CI/CD pipelines, and driving DevOps best practices across hybrid and cloud-native environments. Strong command of **Terraform**, **PowerShell**, and various scripting tools to deliver scalable, secure, and cost-effective solutions.

Extensive hands-on experience managing enterprise-grade workloads, improving operational efficiency, and ensuring high availability and compliance in industries including **financial services and payment systems**. Collaborative leader with a proven ability to align technical execution with business goals, while fostering automation, agility, and innovation across cross-functional teams.

- **Cloud Platforms:** Expertise in **Microsoft Azure** and **Amazon Web Services (AWS)** cloud environments with a strong focus on automation, scalability, and security.
- **DevOps Tooling:** Hands-on experience with CI/CD tools including **Azure DevOps**, **GitHub Actions**, and **Jenkins** for seamless code deployment and infrastructure delivery.
- **Infrastructure as Code (IaC):** Proficient in using **Terraform** and **YAML** for infrastructure provisioning, configuration, and environment standardization.
- **Scripting & Automation:** Advanced scripting skills in **PowerShell**, **Bash**, and **Python** to automate cloud resource management and operational tasks.
- **Disaster Recovery & Backup:** Designed and implemented robust DR strategies using **Azure Site Recovery** and **Azure Backup**, including automated failover testing.
- **Linux Administration:** Strong foundation in **Linux system administration**, including configuration, patching, monitoring, and performance tuning.
- **Containerization & Orchestration:** Experience deploying containerized applications with **Docker** and managing Kubernetes clusters, especially in **Azure Kubernetes Service (AKS)**.

- **Cloud Governance & Compliance:** Implemented cost control, tagging, and compliance strategies aligned with **SOC 2, HIPAA, and GDPR**.
- **Networking & Security:** In-depth understanding of cloud networking (e.g., VNETs, NSGs, VPNs, IAM), securing access, and maintaining operational integrity.
- **Domain Knowledge:** Experience working in **payment systems environments**, including merchant, processor, and banking platforms.
- **Team Collaboration & Leadership:** Proven ability to work closely with developers, security teams, and IT stakeholders to align infrastructure with business priorities.

TECHNICAL SKILLS

Cloud Technologies	Azure, AWS, GCP
Operating Systems	Ubuntu, CentOS, Red Hat, Linux, Windows
Configuration and Infrastructure Management Tools	Terraform, Ansible, Chef, Puppet, Maven, ANT
Automation and Ticketing Tools	GitLab, GitHub, Jenkins, Bamboo, Jira
Containerization	Docker, Docker Swarm, Kubernetes
Scripting Languages	Python, Ruby, JSON, YAML, Shell, GO
Web Servers	Web Logic, JBoss, Apache Tomcat, Websphere
Monitoring Tools	Splunk, Nagios, Cloud Watch, ELK, App Dynamics
Databases	MySQL, SQL Server, MongoDB, DynamoDB
Repository Management	Jfrog, Nexus, Artifactory
Network Protocols	TCP/IP, DHCP, DNS, SNMP, SMTP, Ethernet, NFS

PROFESSIONAL EXPERIENCE

Client: Oil and Gas, Huston, Tx.

Dec 2022 – Present

Role: Azure DevOps Architect (Migration from On prem to Azure Cloud)

Responsibilities:

Cloud & DevOps (Azure, ADO, ADF, Azure AD-EntraID, Snowflake, Kubernetes, Docker, CI/CD, Terraform):

- Designed and implemented Azure cloud architectures, ensuring security, scalability, and high availability.
- Built and maintained **CI/CD** pipelines in **Azure DevOps** (ADO) for automated deployments and testing.

- Managed **Kubernetes** (AKS) clusters and deployed containerized applications using **Docker**.
- Automated infrastructure provisioning with **Terraform**, ensuring compliance and consistency.
- Integrated **CyberArk Conjur** for secrets management in DevOps workflows and containerized applications.
- Azure log **Monitoring** for configuring and managing log data collection.
- Monitor application performance and diagnostics in application insights. And look for exceptions for web applications.
- Alerting and automation by setting up alerts and integrating with **Azure logic apps**, power automate, Azure functions.
- Strong expertise in **Snowflake Data Cloud**, including data warehousing, performance tuning, and query optimization.
- Experience in **Snowflake schema design**, data modeling, and best practices for high-performance analytics.
- Proficient in **Snowflake SQL**, stored procedures, and Snowflake Python Connector for **ETL/ELT** data pipelines.
- Added **IP addresses** in **Network** in **Azure** for applications in all three environments for multiple locations.

Security & Compliance (OneTrust, GRC, Risk Assessment, Enterprise Architecture, MS Visio, CyberArk, Conjur, ServiceNow, PowerBI, Citrix workspace):

- Implemented **CyberArk** Privileged Access Management (PAM) and Conjur for secure credential management.
- Created usernames and **passwords** for web apps and stored in conjur for Azure web apps.
- Stored client secrets in **Conjur** and sync with Azure so that **client secret** certificate is rotated itself and doesn't expire. Also sync itself if a new secret is created in azure.
- Conducted **risk assessments** and security reviews for cloud workloads, DevOps processes, and .NET and Mendix web applications.
- Created .NET web apps architectures and Mendix web apps architectures for **Enterprise Architecture** approvals, Security approvals and **GRC** approvals.
- Managed GRC (**Governance, Risk, and Compliance**) approvals for security exceptions and policy adherence.
- Automated security and risk workflows using **ServiceNow** GRC module, ensuring efficient tracking and compliance reporting.
- Applied security best practices in Azure IAM, RBAC, network security groups (NSGs), and firewall policies.
- Updated CI relationships, Descriptions, Owners in ServiceNow for Business applications and Web applications.
- Raised **CMDB** requests to update the web apps url's in ServiceNow

Architecture & Database Management (Architectures, Sharepoint, SQL, Teradata Studio, Terraform, Visual Studio, Kubernetes, Docker):

- Designed and optimized secure cloud and **on-prem architectures**, aligning with enterprise security policies.
- Developed and optimized **SQL queries**, stored procedures, and database performance tuning.
- Managed containerized database services with **Docker and Kubernetes**, ensuring high availability.

- Managed and optimized **Teradata** databases using Teradata Studio for data analysis and reporting. Further **monitoring logs** in azure portal for errors and **exceptions**.
- Automated infrastructure and database provisioning using Terraform, improving deployment efficiency.
- Documented the .NET web apps and Mendix web apps along with creating their **architectures** in folders in Sharepoint to make easy access for maintenance team after team is rolled-off the project.
- Integration of **Snowflake** with **Azure Data Factory**, **Databricks**, and **Power BI** for data transformation and reporting.

Along with:

- **GenAI Integration in Operations:** Utilized **Generative AI** to optimize predictive maintenance by analyzing sensor and operational data, improving asset management and reducing downtime.
- Integrated AI-driven insights into **Exploration and Production (E&P)** processes, optimizing reservoir management, and identifying drilling opportunities.
- **Process Optimization & Workflow Automation:** Leveraged GenAI to streamline **supply chain optimization**, reducing operational costs through predictive demand forecasting and logistics automation.
- Implemented AI models for real-time monitoring of safety parameters and risk management, ensuring **compliance** and worker safety.
- **Data-Driven Decision-Making:** Developed **data analysis pipelines** to process historical and real-time data for predictive modeling, improving decision-making and operational efficiency.
- Enhanced decision-making through AI-driven insights for **energy trading**, forecasting market trends, and developing trading strategies.
- **Compliance & Reporting Automation:** Automated regulatory reporting and compliance documentation generation using **GenAI**, ensuring accurate and timely submission in line with industry standards.
- **Collaboration & Continuous Improvement:** Collaborated with cross-functional teams to integrate AI tools into **DevOps workflows**, driving continuous improvement and operational agility.
- **AI & Application Modernization:** Design, develop, and deploy AI-powered software solutions using Azure AI Services to enhance customer experiences.
- Integrate Azure AI capabilities (Search, Vision, Speech, and Document Intelligence) into enterprise applications.
- Implement large language models (LLMs) and vector databases for AI-driven functionalities.
- Optimize cost and resource allocation of AI models in Azure.
- Ensure AI systems comply with security, data governance, and ethical AI guidelines.

Client: Sirius, San Antonio, TX

Dec 2020 – Dec 2022

Role: Sr Azure DevOps Engineer (Hosted Apps on Azure and GCP Cloud)

Responsibilities:

- Designed and enhanced scalable, reliable, and high-performance **automated build and test infrastructure**, improving development velocity and reducing time to deployment.
- Streamlined software delivery lifecycle (SDLC) by optimizing integration and deployment pipelines for **Development** and **Software QA teams**, incorporating AI-based recommendations and validation checks.

- Led continuous refinement of deployment workflows to reduce downtime and deployment risk across **microservices-based enterprise applications** hosted on **Azure** and **GCP**.
- Administered and optimized cloud environments across **Microsoft Azure** and **Google Cloud Platform (GCP)**, including infrastructure provisioning, performance tuning, and workload automation.
- Implemented **Infrastructure as Code (IaC)** using **Bicep** and **Terraform**, with Bicep as the preferred tool for declarative resource management and pipeline integration.
- Managed application hosting and orchestration using **Docker**, **Kubernetes**, and **Helm**, ensuring seamless deployments, auto-scaling, and high availability of containerized services.
- Developed end-to-end automation solutions using tools such as **Ansible**, **Puppet**, and **Chef**, enhancing repeatability and consistency across staging and production environments.
- Integrated observability tools including **Grafana**, **Prometheus**, and **Elasticsearch** for real-time performance monitoring, health checks, and anomaly detection across critical services.
- Implemented continuous delivery and blue-green deployments using **Jenkins**, **Terraform**, and container-based environments, aligned with enterprise DevSecOps standards.
- Provided deep expertise in **Linux administration**, **shell scripting**, and deployment automation, supporting internal tooling and large-scale cloud-native platforms.
- Supported virtualization environments via **VMware**, **KVM**, and **libvirt**, optimizing VM provisioning and workload placement strategies for dev/test infrastructure.
- Integrated **RabbitMQ** messaging queues to support resilient and scalable microservices communication patterns.
- Collaborated with developers, QA engineers, and release managers to ensure efficient integration of new features into CI/CD pipelines and production environments.
- Maintained clear and comprehensive documentation for deployment workflows, configuration standards, rollback procedures, and automation runbooks.
- Actively researched emerging trends in **DevOps**, **Cloud-Native tooling**, **automation**, applying best practices to improve platform performance and resilience.
- Contributed to cloud-to-cloud migration projects, optimizing compute and storage efficiency during transitions between Azure and GCP.
- Familiarity with **Microsoft Digital Twins**, enabling simulation-driven infrastructure testing and digital modeling for advanced use cases.
- Experience with **Node.js** in DevOps contexts, supporting back-end microservices and scripting solutions for automation tasks.

Client: Crowe, Austin, TX

Jan 2020 – Nov 2020

Role: Azure DevOps Engineer

Responsibilities:

- Designed and implemented resilient DR strategies using **Azure Site Recovery (ASR)** and **Azure Backup**, aligning with RTO and RPO targets.

- Performed automated **failover testing** and recovery drills to ensure readiness for high-availability scenarios.
- Developed and maintained **Terraform modules** for provisioning and managing scalable Azure infrastructure.
- Automated infrastructure deployments using **PowerShell**, **YAML**, and CI/CD pipelines in **Azure DevOps**, **GitHub Actions**, and **Jenkins**.
- Ensured secure state management and consistent deployments across development, staging, and production environments.
- Engineered end-to-end automation scripts in **PowerShell**, **Bash**, and **Python** to manage Azure resources and streamline operational tasks.
- Integrated automation into CI/CD processes to support infrastructure scaling, monitoring, and configuration management
- Provisioned and managed Azure services, including **VMs**, **VNETs**, **Load Balancers**, **Storage Accounts**, and **IAM** roles.
- Architected secure hybrid cloud networks with **VPN Gateways**, **NSGs**, and **ExpressRoute**.
- Deployed monitoring solutions using **Azure Monitor**, **Log Analytics**, and **Application Insights**.
- Implemented containerized applications using **Docker** and deployed to **Azure Kubernetes Service (AKS)** environments.
- Integrated Kubernetes with CI/CD pipelines for seamless application delivery.
- Applied tagging strategies and cost management techniques to optimize cloud spend and track resource utilization.
- Ensured infrastructure compliance with standards such as **SOC 2**, **HIPAA**, and **GDPR**.
- Partnered with cross-functional teams including developers, IT operations, and security to align infrastructure goals.
- Authored technical documentation for DR procedures, Terraform modules, and CI/CD workflows for team adoption.
- Brought domain-specific insight to cloud infrastructure design in **payment systems**, including merchant, provider/processor, and financial institution contexts.

Client: Credit Karma, San Francisco, CA.

May 2017 – Dec 2019

Role: Sr AWS DevOps Engineer

Responsibilities:

- **Architected and managed scalable cloud infrastructure on AWS**, leveraging core services such as EC2, S3, RDS, Lambda, IAM, and VPC to support highly available, fault-tolerant applications.
- **Implemented Infrastructure as Code (IaC)** using **Terraform** and **AWS CloudFormation**, enabling repeatable, consistent environment provisioning and improving deployment efficiency.
- **Built and maintained CI/CD pipelines** using **Jenkins**, **GitLab CI**, and **CircleCI**, automating application builds, tests, and deployments to reduce cycle time and increase release velocity.

- **Developed serverless applications** utilizing **AWS Lambda** functions written in **Python**, **JavaScript (Node.js)**, and **Go**, integrating with services such as API Gateway and DynamoDB for modern, cost-effective backend solutions.
- **Applied GitOps principles** by designing and enforcing Git-centric workflows (feature branching, pull requests, protected branches) and maintaining clean, audit-friendly version control using **GitHub** or **GitLab**.
- **Integrated code quality and security tools** into development pipelines, including static analysis (e.g., SonarQube), linters (e.g., ESLint, Pylint), and open-source vulnerability scanning tools.
- **Led containerization efforts** using **Docker**, and contributed to **Kubernetes** or **AWS ECS/EKS** cluster configuration, improving microservices deployment and scaling capabilities.
- **Collaborated across cross-functional teams**, presenting technical concepts to both technical and non-technical stakeholders, and translating business requirements into scalable technical solutions.
- **Mentored junior engineers**, conducting code reviews, sharing DevOps best practices, and promoting a culture of continuous improvement.
- **Demonstrated strong project management and time prioritization skills** in fast-paced, agile environments, effectively handling multiple priorities and stakeholder demands.
- **Represented engineering in customer-facing engagements**, including architecture consultations, technical presentations, and troubleshooting sessions, ensuring alignment between business and technical goals.

Client: Verizon, Cary, NC

Mar 2014 – April 2017

Role: AWS DevOps Infrastructure Engineer

Responsibilities:

- **Designed and built scalable, high-performance backend systems** to support core Customer Data Platform (CDP) capabilities, ensuring long-term reliability and maintainability in a distributed cloud environment (AWS).
- **Refactored legacy services and addressed technical debt** by applying modern engineering best practices, improving system performance, and increasing overall code quality.
- **Maintained platform stability and operational excellence** through proactive monitoring, root cause analysis, and active participation in the platform's on-call rotation.
- **Collaborated with cross-functional teams** including Product, Frontend, and DevOps, to align backend infrastructure with user needs and product goals, fostering a strong culture of partnership and knowledge sharing.
- **Mentored junior engineers and peers**, promoting technical growth and best practices through code reviews, architectural discussions, and internal documentation.
- **Drove key strategic initiatives** such as Core CDP Infrastructure Consolidation, contributing to long-term architectural planning and platform unification.
- **Enhanced CI/CD processes and pipeline reliability** by implementing and optimizing workflows using Jenkins and Infrastructure as Code tools such as Terraform and Ansible.

- **Implemented robust observability solutions** using tools like Datadog, Prometheus, and New Relic to ensure visibility into system health and improve incident response.
- **Applied networking and infrastructure expertise** (e.g., TCP/IP, Load Balancing, DNS, OSI Model) to architect resilient backend services across distributed systems.
- **Promoted a security-first development approach**, integrating tools like Snyk and SonarQube for continuous vulnerability scanning and code quality assurance.

Client Axis Bank, Punjab, India

Jan 2013 – Mar 2014

Role: **LINUX SYSTEM ADMINISTRATOR**

Responsibilities:

- **Server Administration & Maintenance:** Install, configure, and maintain Linux distributions such as RHEL, CentOS, Ubuntu, Debian, and SUSE.
- Perform kernel upgrades and patches to enhance system performance and security.
- Monitor system health, performance, and resource utilization using tools like Nagios, Zabbix, and Munin.
- **User & Access Management:** Manage user accounts, groups, and permissions using LDAP, NIS, and local authentication.
- Implement Role-Based Access Control (RBAC) and manage SSH access.
- **Filesystem & Storage Management:** Configure and manage LVM (Logical Volume Manager), RAID, and disk partitions.
- Implement file system quotas and permissions to ensure security and fair resource allocation.
- Perform regular data backups and disaster recovery planning using rsync, tar, and Bacula.
- **Networking & Security:** Configure and manage firewalls using iptables and firewall.
- Troubleshoot and optimize network performance using Wireshark, tcpdump, and netstat.
- Implement secure remote access via OpenSSH, OpenVPN, and IPSec.
- **Automation & Scripting:** Automate repetitive system administration tasks using Bash, Perl, and Python scripts.
- Manage system configurations and automation with Puppet and Chef.
- **Server & Application Management:** Deploy and maintain Apache, Nginx, and Tomcat web servers.
- Optimize database performance and perform basic DBA tasks for MySQL and PostgreSQL.